

YANDING MARIO YIN SOVENIS

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PROJECTS

Assistant Lead Frontend Engineer, collaboration with Brigham & Woman's Hospital **March - May 2025**

- Developed a full-stack prototype website for Massachusetts General Brigham hospital in 10 weeks as part of a 10 person team. Deployed with Prisma ORM on AWS EC2 with Docker for containerization.
- Led frontend development using TypeScript, React, Mantine for UI components and Github for PR quality control.
- Directed a team of 4 engineers following Agile methodologies, daily scrums and software design patterns across 4 sprints.
- Collected 30+ requirements by creating user stories and scenarios.
- Integrated Google Maps APIs, text-to-speech and speech-to-text functionalities, enhancing the user experience.

Full Stack Software Engineer, collaboration with WPI CS department **November - January 2026**

- Partnered with Doctoral Faculty to develop a full-stack spreadsheet tool application currently integrated as the primary instructional tool for 50+ students in advanced CS courses.
- Refactored a legacy JavaFX interface into a decoupled React architecture, as well as implemented memoization strategies using React Hooks for optimized rendering.
- Designed an event-driven Backend using Springboot REST APIs, integrating the Observer Design Pattern to map complex cell dependencies and trigger cascading logic updates between the backend and React frontend.

Full Stack Software Engineer, Restaurant WebApp **February - April 2024**

- Engineered a full-stack WebApp featuring a QR code-based ordering system deployed in restaurants. Developed by a 6-person team following agile methodologies and daily scrums throughout the software development lifecycle.
- Designed the application using Node.js, React.js, Javascript, Express.js, and Bootstrap.
- Directed backend implementation of RESTful APIs for scalable dynamic menu management, real-time order tracking, and ESC/POS-integrated billing system, ensuring communication between services.

Machine Learning Study, Collaboration with Hangzhou Dianzi University **October - December 2024**

- Collaborated with ML engineers to evaluate and benchmark advanced models for urban traffic flow prediction, identifying strengths and limitations for each and their practical applications in public transportation planning and traffic congestion detection.
- Conducted a comparative analysis of neural network architectures, including Graph Convolutional Networks, Gated Recurrent Units, and the hybrid Time-Graph Convolutional Network, as well as traditional linear models.
- Processed real-world traffic datasets from Hangzhou and Shenzhen using Pandas and NumPy, preparing adjacency matrices and feature matrices for spatio-temporal analysis.
- Evaluated model performance across multiple forecasting horizons (15, 30, 45, and 60 minutes) using key metrics such as Root Mean Squared Error, Mean Absolute Error, and Prediction Accuracy.

SKILLS

- **Programming Languages:** Python, Javascript, Java, Typescript, C, C++, SQL, Bash, MATLAB
- **Frameworks/Libraries:** Node.js, Express, React, Next.js, Bootstrap, Prisma, Mantine, Linux, Pytorch, Tensorflow
- **Development Tools:** Git, Docker, Postman, AWS, VSCode, Webstorm, Wireshark, Jira, Vim
- **Languages:** English, Spanish, Chinese

ADDITIONAL EXPERIENCE

Student Teaching Assistant, Computer Science Department (WPI) **August 2024 - Present**

- Delivered one on one office hours sessions to help students understand programming concepts.
- Assessed exams, graded projects, conducted lab sessions across CS Courses.

National Math Competition, Ecuador **June 2021**

- Bronze Medal

EDUCATION

Worcester Polytechnic Institute (WPI) Worcester, MA **GPA: 3.7 | Expected May 2026**

- B.S in Computer Science
- Minor in Chinese
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Relevant Coursework: Operating Systems, Object Oriented Design, Computer Networks, Data Structures/Algorithms, Database Systems, Software Engineering, Machine Learning, Applied NLP